Economics 880 - PS4

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Question 1

The moment vector $m_3(x_t)$ contains the quantities used to construct the mean, variance, and first-order autocovariance of the time series $x \equiv \{x_t\}$. We have

$$m_3(x_t) = \begin{bmatrix} x_t \\ (x_t - \bar{x})^2 \\ (x_t - \bar{x})(x_{t-1} - \bar{x}) \end{bmatrix}.$$

We will consider the moments of the stationary distribution of the series since we are asked to compute the asymptotic moments. We have

$$\begin{aligned} \mathbb{E}[x_t] &= \rho_0 \mathbb{E}[x_{t-1}] + \mathbb{E}[\varepsilon_t] \\ &= \rho_0 \mathbb{E}[x_{t-1}]. \end{aligned}$$

Under stationarity, $\mathbb{E}[x_t] = \mu$ for all t, hence

$$\mu = \rho_0 \mu \qquad \Rightarrow \qquad \mu(1 - \rho_0) = 0 \qquad \Rightarrow \qquad \mu = 0,$$

since $\rho_0 = 0.5$.

For the asymptotic variance, we have

$$\begin{aligned} \mathbb{E}[x_t^2] &= \rho_0^2 \mathbb{E}[x_{t-1}^2] + 2\rho_0 \mathbb{E}[x_{t-1}\varepsilon_t] + \mathbb{E}[\varepsilon_t^2] \\ &= \rho_0^2 \mathbb{E}[x_{t-1}^2] + \sigma_0^2, \end{aligned}$$

since $\mathbb{E}[x_{t-1}\varepsilon_t] = 0$. Under stationarity, $\mathbb{E}[x_{t-1}^2] = \mathbb{E}[x_t^2] = \nu$, so

$$\nu = \rho_0^2 \nu + \sigma_0^2 \qquad \Rightarrow \qquad \nu = \frac{\sigma_0^2}{1 - \rho_0^2}.$$

For the first-order autocovariance, we have

$$\begin{aligned} \mathbb{E}[x_t x_{t-1}] &= \mathbb{E}[(\rho_0 x_{t-1} + \varepsilon_t) x_{t-1}] \\ &= \rho_0 \mathbb{E}[x_{t-1}^2] + \mathbb{E}[\varepsilon_t x_{t-1}] \\ &= \rho_0 \mathbb{E}[x_{t-1}^2] \\ &= \frac{\rho_0 \sigma_0^2}{1 - \rho_0^2}, \end{aligned}$$

where the third equality follows from the independence of ε_t and x_{t-1}, and the fourth from stationarity.

Thus, for the parameter vector $b = (\rho, \sigma^2)^\top$, the population moment vector is

$$\mu_3(b) = \begin{bmatrix} 0 \\ \dfrac{\sigma^2}{1-\rho^2} \\ \dfrac{\rho\sigma^2}{1-\rho^2} \end{bmatrix}.$$

Evaluating at the true parameters, $b_0 = (\rho_0, \sigma_0^2)^\top$, gives

$$\mu_3(b_0) = \begin{bmatrix} 0 \\ \dfrac{\sigma_0^2}{1-\rho_0^2} \\ \dfrac{\rho_0\sigma_0^2}{1-\rho_0^2} \end{bmatrix} = \begin{bmatrix} 0 \\ 4/3 \\ 2/3 \end{bmatrix}.$$

We next compute the Jacobian of the population moment conditions. Recall that

$$g(b) = \mu_3(b_0) - \mu_3(b),$$

so

$$\nabla_b g(b) = -\frac{\partial \mu_3(b)}{\partial b^\top}.$$

Differentiating with respect to ρ and σ^2, we obtain

$$\nabla_b g(b) = -\begin{bmatrix} 0 & 0 \\ \dfrac{2\rho\sigma^2}{(1-\rho^2)^2} & \dfrac{1}{1-\rho^2} \\ \dfrac{\sigma^2(1+\rho^2)}{(1-\rho^2)^2} & \dfrac{\rho}{1-\rho^2} \end{bmatrix}.$$

Evaluating at b_0, we have

$$G \equiv \nabla_b g(b_0) = -\begin{bmatrix} 0 & 0 \\ 16/9 & 4/3 \\ 20/9 & 2/3 \end{bmatrix}.$$

The Jacobian has full column rank since

$$\det \begin{bmatrix} 16/9 & 4/3 \\ 20/9 & 2/3 \end{bmatrix} = -\frac{16}{9} \neq 0.$$

Thus,

$$\operatorname{rank}(G) = 2,$$

and the two structural parameters are locally identified.

We next compute the AGS sensitivity matrix, defined as

$$\Lambda = -(G^T W G)^{-1} G^T W.$$

Using $W = I_3$, we have

$$\Lambda = -(G^T G)^{-1} G^T.$$

Substituting the Jacobian and simplifying gives

$$\Lambda = \begin{bmatrix} 0 & -3/8 & 3/4 \\ 0 & 5/4 & -1 \end{bmatrix}.$$

The first column of Λ is zero, indicating that the sample mean does not affect either parameter estimate when $W = I_3$. This follows from the fact that the population mean is zero regardless of the structural parameters.

The variance and first-order autocovariance are informative for both parameters. In particular, the first row of Λ indicates that an increase in the empirical variance decreases the estimate of ρ, holding the autocovariance fixed, while an increase in the empirical autocovariance increases the estimate of ρ, holding the variance fixed.

Similarly, the second row indicates that an increase in the empirical variance increases the estimate of σ^2, holding the autocovariance fixed, while an increase in the empirical autocovariance decreases the estimate of σ^2, holding the variance fixed.

Thus, the mean provides no identifying information, while the variance and autocovariance jointly identify the two structural parameters.

Questions 4–6

In the numerical implementation, we parameterize the model by

$$b = (\rho, \sigma)^T,$$

where σ denotes the standard deviation of the innovations, rather than their variance as in Question 1.

For a given set of moments, the SMM objective is

$$J_{TH}(b) = g_{TH}(b)^T W g_{TH}(b),$$

where

$$g_{TH}(b) = M_T(x) - M_{TH}(y(b)).$$

We first obtain $\widehat{b}_1$ using $W_1 = I$. We then construct the Newey–West covariance estimator with four lags,

$$\widehat{S}_{TH} = \left(1 + \frac{1}{H}\right) \widehat{S}_{y,TH},$$

and reestimate the parameters using

$$W_2 = \widehat{S}_{TH}^{-1}.$$

The resulting estimator is denoted $\widehat{b}_2$.

For inference, we approximate the Jacobian $G = \nabla_b g_{TH}(\widehat{b}_2)$ numerically and calculate

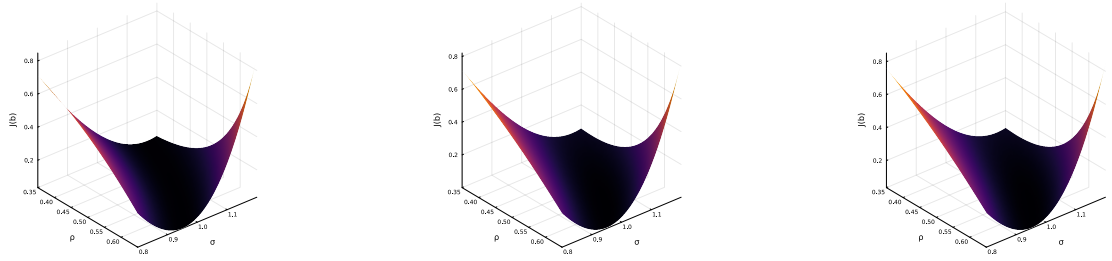
$$\widehat{\mathrm{Var}}(\widehat{b}_2) = \frac{1}{T}(G^T W_2 G)^{-1},$$

along with the AGS sensitivity matrix

$$\Lambda = -(G^T W_2 G)^{-1} G^T W_2.$$

Because the weighting matrix already incorporates the $(1+1/H)$ simulation adjustment, the J-test statistic is

$$J_{test} = T J_{TH}(\widehat{b}_2).$$



(a) Mean and variance (b) Variance and autocorrelation (c) All three moments

Figure 1: First-stage SMM objective functions over $\rho \in [0.35, 0.65]$ and $\sigma \in [0.8, 1.2]$. All three specifications include the variance moment, which accounts for much of the similarity between the objective surfaces.

Question 4

We first consider the just-identified specification matching the mean and variance:

$$M_T(x) = \begin{bmatrix} \bar{x} \\ \widehat{\mathrm{Var}}(x) \end{bmatrix} = \begin{bmatrix} 0.217749 \\ 1.533191 \end{bmatrix}.$$

From Question 1, the corresponding population moments are

$$\mu_4(b) = \begin{bmatrix} 0 \\ \dfrac{\sigma^2}{1-\rho^2} \end{bmatrix}.$$

Their Jacobian at the true parameters, using the parameterization $b = (\rho, \sigma)^\top$, is

$$G_4(b_0) = -\begin{bmatrix} 0 & 0 \\ 16/9 & 8/3 \end{bmatrix}.$$

The Jacobian has rank one. The mean provides no identifying information, while the variance identifies only the combination $\sigma^2/(1-\rho^2)$. Consequently, ρ and σ are not separately identified.

(a) First-stage estimation. Minimizing the SMM objective with $W_1 = I_2$ gives

$$\widehat{b}_1 = \begin{bmatrix} 0.998306 \\ 0.208981 \end{bmatrix}, \qquad J_{TH}(\widehat{b}_1) = 0.006651.$$

Figure 1(a) presents the objective surface over the specified parameter grid. The estimated parameters lie outside this grid, with the autoregressive coefficient approaching the stationarity boundary.

This behavior is consistent with the identification problem: different combinations of ρ and σ can generate similar unconditional variances.

(b) Second-stage estimation. Using the Newey–West covariance estimator, we obtain

$$W_2 = \begin{bmatrix} 0.047149 & -0.000057 \\ -0.000057 & 0.029513 \end{bmatrix}.$$

The second-stage optimizer returns

$$\widehat{b}_2 = \begin{bmatrix} 0.998306 \\ 0.208981 \end{bmatrix}, \qquad J_{TH}(\widehat{b}_2) = 0.000313.$$

However, the second-stage optimization did not satisfy its convergence criterion.

(c) Standard errors and AGS statistics. The numerical Jacobian evaluated at the reported second-stage estimates is

$$\widehat{G}_4 = \begin{bmatrix} -3.723984 & -0.652084 \\ -83.932987 & -14.706804 \end{bmatrix}.$$

The resulting parameter covariance matrix is

$$\widehat{\mathrm{Var}}(\widehat{b}_2) = \begin{bmatrix} 17239.161 & -98385.565 \\ -98385.565 & 561495.965 \end{bmatrix},$$

which implies standard errors

$$\widehat{\mathrm{SE}}(\widehat{b}_2) = \begin{bmatrix} 131.298 \\ 749.330 \end{bmatrix}.$$

The AGS sensitivity matrix is

$$\widehat{\Lambda}_4 = \begin{bmatrix} 402.590 & -17.850 \\ -2297.615 & 101.942 \end{bmatrix}.$$

The singular values of the estimated Jacobian are approximately 85.296 and 0.000428. The condition number of $\widehat{G}_4^{\mathsf{T}} W_2 \widehat{G}_4$ is approximately 2.49×10^{10}.

These results indicate severe numerical ill-conditioning, consistent withrank deficiency. The very large standard errors and AGS sensitivities reflect the difficulty of separately estimating the two parameters. Since the Jacobian is rank deficient, the usual asymptotic covariance and AGS formulas are not well-defined at the true parameter values.

(d) J-test. The reported J-statistic is $J_{\text{test}} = 0.062679$.

Question 5

We next consider the just-identified specification matching the variance and first-order autocorrelation:

$$M_T(x) = \begin{bmatrix} 1.533191 \\ 0.563408 \end{bmatrix}.$$

The population moments are

$$\mu_5(b) = \begin{bmatrix} \dfrac{\sigma^2}{1-\rho^2} \\ \rho \end{bmatrix}.$$

The Jacobian at the true parameters is

$$G_5(b_0) = -\begin{bmatrix} 16/9 & 8/3 \\ 1 & 0 \end{bmatrix}.$$

Since

$$\det(G_5(b_0)) = -\frac{8}{3} \neq 0,$$

the Jacobian has full column rank. Unlike the specification in Question 4, these moments locally identify

both parameters. In particular, autocorrelation identifies ρ, while variance then identifies σ.

(a) First-stage estimation. Using $W_1 = I_2$, we obtain

$$\widehat{b}_1 = \begin{bmatrix} 0.576984 \\ 1.021874 \end{bmatrix}, \qquad J_{TH}(\widehat{b}_1) \approx 0.$$

Figure 1(b) displays the objective surface. The parameter estimates are reasonably close to the true values $b_0 = (0.5, 1)^\top$.

(b) Second-stage estimation. The estimated optimal weighting matrix is

$$W_2 = \begin{bmatrix} 0.133942 & -0.215347 \\ -0.215347 & 1.713902 \end{bmatrix}.$$

The second-stage estimates are

$$\widehat{b}_2 = \begin{bmatrix} 0.576984 \\ 1.021874 \end{bmatrix}, \qquad J_{TH}(\widehat{b}_2) \approx 0.$$

(c) Standard errors and AGS statistics. The numerical Jacobian is

$$\widehat{G}_5 = \begin{bmatrix} -2.610561 & -3.000709 \\ -0.985242 & 0.000000 \end{bmatrix}.$$

The estimated parameter covariance matrix is

$$\widehat{\mathrm{Var}}(\widehat{b}_2) = \begin{bmatrix} 0.003766 & -0.001288 \\ -0.001288 & 0.004587 \end{bmatrix},$$

giving standard errors

$$\widehat{\mathrm{SE}}(\widehat{b}_2) = \begin{bmatrix} 0.061369 \\ 0.067724 \end{bmatrix}.$$

The AGS sensitivity matrix is

$$\widehat{\Lambda}_5 = \begin{bmatrix} 0.000000 & 1.014979 \\ 0.333255 & -0.883013 \end{bmatrix}.$$

The columns correspond to variance and autocorrelation, respectively, and the rows to ρ and σ.

The first row indicates that the estimate of ρ is almost entirely sensitive to autocorrelation, as predicted by the population moments. The estimate of σ responds positively to variance and negatively to autocorrelation.

The Jacobian has singular values 4.0314 and 0.7333. Furthermore, the condition number of $\widehat{G}_5^\top W_2 \widehat{G}_5$ is approximately 1.96.

These results are consistent with local identification, and the standard errors are substantially smaller than those obtained in Question 4.

(d) J-test. The J-statistic is approximately $J_{\text{test}} = 7.10 \times 10^{-9}$.

Question 6

Finally, we consider the overidentified specification matching the mean, variance, and first-order autocorrelation:

$$M_T(x) = \begin{bmatrix} 0.217749 \\ 1.533191 \\ 0.563408 \end{bmatrix}.$$

The corresponding population moment vector is

$$\mu_6(b) = \begin{bmatrix} 0 \\ \dfrac{\sigma^2}{1-\rho^2} \\ \rho \end{bmatrix},$$

with Jacobian

$$G_6(b_0) = -\begin{bmatrix} 0 & 0 \\ 16/9 & 8/3 \\ 1 & 0 \end{bmatrix}.$$

The Jacobian has full column rank, so the two parameters are locally identified. Since we have three moments and two parameters, the model is overidentified.

(a) First-stage estimation. Using the identity weighting matrix, we obtain

$$\widehat{b}_1 = \begin{bmatrix} 0.586628 \\ 1.013961 \end{bmatrix}, \qquad J_{TH}(\widehat{b}_1) = 0.035732.$$

Figure 1(c) presents the objective surface. Unlike Question 5, the three moment conditions cannot in general be matched exactly using only two parameters.

(b) Second-stage estimation. The estimated optimal weighting matrix is

$$W_2 = \begin{bmatrix} 0.227746 & 0.011538 & -0.005027 \\ 0.011538 & 0.132807 & -0.217970 \\ -0.005027 & -0.217970 & 1.747638 \end{bmatrix}.$$

The second-stage estimates are

$$\widehat{b}_2 = \begin{bmatrix} 0.580953 \\ 1.026909 \end{bmatrix}, \qquad J_{TH}(\widehat{b}_2) = 0.008087.$$

The optimal weighting matrix changes the relative importance of the moment discrepancies, leading to slightly different estimates from the first stage.

(c) Standard errors and AGS statistics. The numerical Jacobian is

$$\widehat{G}_6 = \begin{bmatrix} -0.073794 & -0.028155 \\ -2.691042 & -3.036088 \\ -0.985274 & 0.000000 \end{bmatrix}.$$

The resulting parameter covariance matrix is

$$\widehat{\mathrm{Var}}(\widehat{b}_2) = \begin{bmatrix} 0.003698 & -0.001317 \\ -0.001317 & 0.004546 \end{bmatrix},$$

and the standard errors are

$$\widehat{\mathrm{SE}}(\widehat{b}_2) = \begin{bmatrix} 0.060815 \\ 0.067427 \end{bmatrix}.$$

The AGS sensitivity matrix is

$$\widehat{\Lambda}_6 = \begin{bmatrix} 0.020815 & -0.000193 & 1.013914 \\ 0.026379 & 0.329127 & -0.900907 \end{bmatrix}.$$

The columns correspond to the mean, variance, and autocorrelation, respectively.

The estimated sensitivities to the mean are relatively small. This is consistent with the population mean being independent of both structural parameters. The small nonzero sensitivities arise from finite-sample variation and the estimated weighting matrix.

As in Question 5, the estimate of ρ is primarily sensitive to autocorrelation, while σ depends on both variance and autocorrelation.

The Jacobian has singular values 4.1117 and 0.7285. The condition number of $\widehat{G}_6^{\mathsf{T}} W_2 \widehat{G}_6$ is approximately 2.01, indicating substantially better numerical conditioning than in Question 4.

(d) J-test. The J-statistic is $J_{\text{test}} = 1.617454$.

(e) Monte Carlo experiment. To examine the finite-sample behavior of the estimator, we perform $B = 500$ Monte Carlo replications. In each replication, we generate a new observed series and a new set of $H = 10$ simulation innovations, then reestimate the parameters using both stages of SMM.

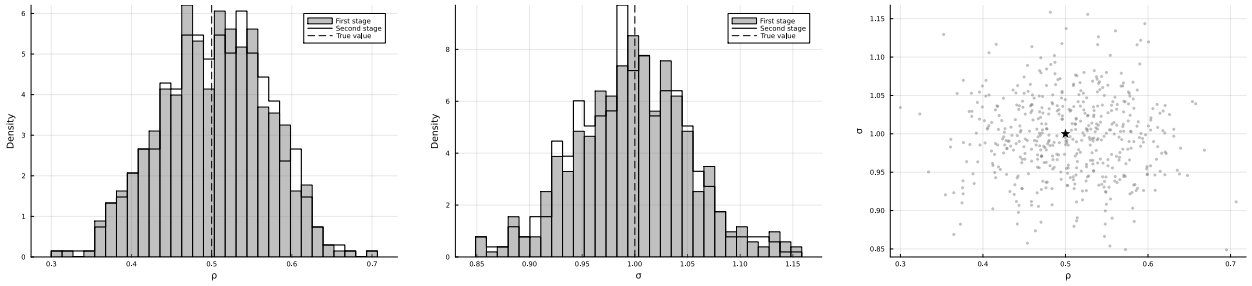
The innovations are fixed across parameter guesses within each replication but independently regenerated across replications.

Table 1 reports the empirical mean, standard deviation, and bias of the parameter estimates.

Table 1: Monte Carlo results for two-stage SMM (B = 500, T = 200, H = 10)

	First stage		Second stage	
	$\widehat{\rho}$	$\widehat{\sigma}$	$\widehat{\rho}$	$\widehat{\sigma}$
True value	0.5000	1.0000	0.5000	1.0000
Mean	0.5027	0.9984	0.5045	0.9964
Standard deviation	0.0665	0.0545	0.0670	0.0540
Bias	0.0027	-0.0016	0.0045	-0.0036

Note: Both optimization stages converged in 496 of 500 replications. The reported statistics include all replications, including the four without reported convergence.



(a) Distribution of $\widehat{\rho}$ (b) Distribution of $\widehat{\sigma}$ (c) Joint second-stage estimates

Figure 2: Monte Carlo distributions of SMM parameter estimates. Panels (a) and (b) compare first- and second-stage estimates, with dashed lines indicating the true parameter values. Panel (c) displays the joint distribution of second-stage estimates, with the true parameter vector marked by a star.

The Monte Carlo means are close to the true parameters in both estimation stages, suggesting relatively small finite-sample bias under the assumed DGP.

The dispersion of the first- and second-stage estimates is also similar. The second-stage estimator has a slightly larger Monte Carlo standard deviation for ρ and a slightly smaller standard deviation for σ. Thus, the optimal weighting matrix does not produce a substantial improvement in finite-sample precision in this particular experiment.

Comparing the Monte Carlo standard deviations with the asymptotic standard errors from part (c), we obtain

	ρ	σ
Asymptotic SE	0.0608	0.0674
Monte Carlo SD	0.0670	0.0540

The asymptotic approximation slightly understates the observed dispersion of $\widehat{\rho}$ and overstates the dispersion of $\widehat{\sigma}$ in this simulation experiment.

Question 7

We estimate the structural parameters of the AR(1) model using indirect inference with an MA(N) auxiliary model for $N \in \{1, 2, 3\}$.

Recall that the stationary AR(1) process admits the MA(∞) representation

$$x_t = \sum_{j=0}^{\infty} \rho^j \varepsilon_{t-j}.$$

We approximate this process using the auxiliary MA(N) model

$$x_t = e_t + \sum_{j=1}^{N} \theta_j e_{t-j}.$$

For each N, we estimate the auxiliary parameters $\widehat{\theta}_N$ by minimizing the sum of squared prediction errors. We additionally calculate the sample standard deviation $\widehat{s}$ to provide information about the scale parameter σ.

For any candidate structural parameter vector $b = (\rho, \sigma)^T$, we simulate $H = 10$ AR(1) series and estimate the auxiliary MA(N) model on each series. We average the resulting auxiliary estimates to obtain $\overline{\theta}_N(b)$ and $\overline{s}(b)$.

We then minimize the indirect inference objective

$$Q_N(b) = \left(\overline{\theta}_N(b) - \widehat{\theta}_N\right)^T \left(\overline{\theta}_N(b) - \widehat{\theta}_N\right) + \left(\overline{s}(b) - \widehat{s}\right)^2.$$

The simulated innovations are held fixed across candidate parameter values during optimization.

Results. The empirical auxiliary parameter estimates are

$$\widehat{\theta}_1 = \begin{bmatrix} 0.460187 \end{bmatrix}, \qquad \widehat{\theta}_2 = \begin{bmatrix} 0.508388 \\ 0.360174 \end{bmatrix},$$

and

$$\widehat{\theta}_3 = \begin{bmatrix} 0.581023 \\ 0.421712 \\ 0.160500 \end{bmatrix}.$$

The empirical sample standard deviation is $\widehat{s} = 1.241328$.

Table 2 reports the estimated structural parameters for each auxiliary specification.

Table 2: Indirect inference estimates using MA(N) auxiliary models

	True value	MA(1)	MA(2)	MA(3)
ρ	0.5000	0.5660	0.5972	0.6311
σ	1.0000	1.0352	1.0079	0.9759
Objective	—	0.000000	0.009781	0.002542

For the MA(1) specification, the estimated structural parameters are

$$\widehat{b}_{\mathrm{MA}(1)} = \begin{bmatrix} 0.565954 \\ 1.035154 \end{bmatrix}.$$

The objective is approximately zero, indicating that the two auxiliary statistics are matched almost exactly. This is consistent with the just-identified nature of the MA(1) specification, which has two auxiliary statistics and two structural parameters.

For the MA(2) specification, we obtain

$$\widehat{b}_{\mathrm{MA}(2)} = \begin{bmatrix} 0.597157 \\ 1.007903 \end{bmatrix},$$

with objective value 0.009781. The estimate of σ is closer to its true value, although the estimate of ρ is farther away.

Finally, for MA(3), we obtain

$$\widehat{b}_{\mathrm{MA}(3)} = \begin{bmatrix} 0.631148 \\ 0.975894 \end{bmatrix},$$

with objective value 0.002542. The estimated persistence parameter is farther from its true value than under either lower-order auxiliary model.